Chapter 2

Limits and Their Duals

2.1 Terminal and Initial Objects

Definition 2.1.1 (Terminal Object)**.** A *terminal object* 1 of a category $\mathcal{C}$ is an object such that for any object A, there is a unique morphism $!_A : A \longrightarrow 1$.

A category may have any number of terminal objects.

Theorem 2.1.2. A terminal object is unique up to isomorphism.

Proof. Suppose that 1 and $1'$ are terminal objects in a category $\mathcal{C}$. On one hand, since 1 is a terminal object, there is a unique morphism $!_{1'} : 1' \longrightarrow 1$. On the other hand, since $1'$ is a terminal object, there is a unique morphism $!_1 : 1 \longrightarrow 1'$.

$$1' \underset{!_1}{\overset{!_{1'}}{\rightleftarrows}} 1$$

Then the composite $!_{1'} \circ !_1 : 1 \longrightarrow 1' \longrightarrow 1$ equals id_1, since the morphism to 1 is unique. In the same way, the composite $!_1 \circ !_{1'} : 1' \longrightarrow 1 \longrightarrow 1'$ equals $id_{1'}$, since the morphism to $1'$ is also unique. Therefore, $!_1$ and $!_{1'}$ are mutually inverse and $1 \cong 1'$. $\square$

Definition 2.1.3 (Global Element)**.** A *global element* of an object A in a category $\mathcal{C}$ is any morphism $1 \longrightarrow A$.

Definition 2.1.4 (Initial Object)**.** An *initial object* $\emptyset$ of a category $\mathcal{C}$ is an object such that for any object C, there is a unique morphism $!_C : \emptyset \longrightarrow C$.

A category may have any number of initial objects.

Theorem 2.1.5. An initial object is unique up to isomorphism.

Proof. By the dual of the proof of Theorem 2.1.2. □

Theorem 2.1.6. For any object A and morphism $f : A \longrightarrow \emptyset$, f is split epic.

Proof. Consider a morphism $f \circ !_A : \emptyset \longrightarrow A \longrightarrow \emptyset$. Since there is only one morphism from $\emptyset$ to any object, one has $f \circ !_A = id_\emptyset$. Thus f has a right inverse $!_A : \emptyset \longrightarrow A$ and is split epic.

$$\begin{tikzcd}
& A \arrow[dl, "f"'] & \\
\emptyset & & \emptyset \arrow[ll, "id_\emptyset"] \arrow[ul, "!_A"']
\end{tikzcd}$$

□

Exercise 2.1.7. Suppose that there is a terminal object 1. Show that every arrow $f : 1 \longrightarrow B$ is monic.

Exercise 2.1.8. An object which is both terminal and initial is called a *zero object*. Show that, if there is an initial object $\emptyset$, a terminal object 1, and an arrow $1 \longrightarrow \emptyset$, then 1 and $\emptyset$ are zero objects.

2.2 Products and Coproducts

Definition 2.2.1 (Binary Product)**.** A *binary product* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A \times B$ of $\mathcal{C}$
- two *projections*: morphisms $\pi_1 : A \times B \longrightarrow A$ and $\pi_2 : A \times B \longrightarrow B$, such that given any object C and morphisms $f : C \longrightarrow A$, $g : C \longrightarrow B$, there is a unique morphism $\langle f, g \rangle : C \longrightarrow A \times B$ for which $\pi_1 \circ \langle f, g \rangle = f$ and $\pi_2 \circ \langle f, g \rangle = g$, i.e. the following diagram commutes.

$$\begin{tikzcd} & C \arrow[dl, "f"'] \arrow[d, dotted, "\langle f,g \rangle"] \arrow[dr, "g"] & \\ A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B \end{tikzcd}$$

We shall simply refer to a binary product $A \times B$ instead of the triple $\langle A \times B, \pi_1, \pi_2 \rangle$.

Exercise 2.2.2. Show that $\langle f, g \rangle$ is monic if f is monic.

Theorem 2.2.3. For any objects A and B, a binary product of A and B is unique up to isomorphism if it exists.

Proof. Suppose that P and P' are binary products of A and B. By Definition 2.2.1, there is a unique arrow $f : P \longrightarrow P'$ such that $\pi_1' \circ f = \pi_1$ and $\pi_2' \circ f = \pi_2$, and there is a unique arrow $g : P' \longrightarrow P$ such that $\pi_1 \circ g = \pi_1'$ and $\pi_2 \circ g = \pi_2'$.

$$\begin{tikzcd} & P \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] \arrow[dd, dotted, bend left, "f"] & \\ A & & B \\ & P' \arrow[ul, "\pi_1'"] \arrow[ur, "\pi_2'"'] \arrow[uu, dotted, bend left, "g"] & \end{tikzcd}$$

Therefore, the composite $g \circ f$ is a morphism from P to P that makes the left diagram below commute, namely, $\pi_1 \circ (g \circ f) = (\pi_1 \circ g) \circ f = \pi_1' \circ f = \pi_1$ and $\pi_2 \circ (g \circ f) = (\pi_2 \circ g) \circ f = \pi_2' \circ f = \pi_2$.

$$
\begin{tikzcd}
 & P \arrow[d, dotted, "f"] \arrow[ddl, "\pi_1"'] \arrow[ddr, "\pi_2"] & \\
 & P' \arrow[d, dotted, "g"] \arrow[dl, "\pi_1'"] \arrow[dr, "\pi_2'"'] & \\
A & P \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}
\qquad
\begin{tikzcd}
 & P \arrow[d, dotted, "id_P"] \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\
A & P \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}
$$

On the other hand, the right diagram above commutes as well. Since P is a binary product of A and B, $g \circ f = id_P$. Similarly, $f \circ g = id_{P'}$. Thus, g and f are mutually inverse, and $P \cong P'$. $\square$

Theorem 2.2.4. For any objects A, B, $A \times B \cong B \times A$.

Proof. Both $\langle A \times B, \pi_1, \pi_2 \rangle$ and $\langle B \times A, \pi_2, \pi_1 \rangle$ are binary products of A and B. Therefore, by Theorem 2.2.3, one has $A \times B \cong B \times A$. $\square$

Exercise 2.2.5. Show that the product constructions are associative: for any objects A, B, C, $A \times (B \times C) \cong (A \times B) \times C$

Definition 2.2.6 (Product of Morphisms)**.** Let $\mathcal{C}$ be a category with finite products and take morphisms $f : A \longrightarrow B$ and $f' : A' \longrightarrow B'$. We write $f \times f' : A \times A' \longrightarrow B \times B'$ for the *product of morphisms* $\langle f\pi, f'\pi' \rangle$ where $\pi : A \times A' \longrightarrow A$ and $\pi' : A \times A' \longrightarrow A'$.

$$
\begin{tikzcd}
A \arrow[d, "f"'] & A \times A' \arrow[l, "\pi"'] \arrow[r, "\pi'"] \arrow[d, dashed, "f \times f'"] & A' \arrow[d, "f'"] \\
B & B \times B' \arrow[l, "\pi"] \arrow[r, "\pi'"'] & B'
\end{tikzcd}
$$

Exercise 2.2.7. Show that each of the following equations hold ($f : A \longrightarrow B,\ f' : A \longrightarrow B',\ g : B \longrightarrow C$, and $g' : B' \longrightarrow C'$).

$$\begin{aligned} id_A \times id_{A'} &= id_{A \times A'} \\ (g \times g') \circ (f \times f') &= g \circ f \times g' \circ f' \\ (g \times g') \circ \langle f, f' \rangle &= \langle g \circ f, g' \circ f' \rangle \end{aligned}$$

Theorem 2.2.8. In a category $\mathcal{C}$ with binary products, any object A is isomorphic to $1 \times A$ ($A \cong 1 \times A$).

Proof. Consider any morphism $f : A \longrightarrow A$ such that $id_A = id_A \circ f$, which means $f = id_A$. Thus, id_A is the unique morphism that makes the following diagram commute.

$$\begin{tikzcd} & A \arrow[dl, "!_A"'] \arrow[d, dotted, "id_A"] \arrow[dr, "id_A"] & \\ 1 & A \arrow[l, "!_A"] \arrow[r, "id_A"'] & A \end{tikzcd}$$

Therefore, $\langle A, !_A, id_A \rangle$ is a binary product of 1 and A. By Theorem 2.2.3, $A \cong 1 \times A$. $\square$

Definition 2.2.9 (Graph)**.** Suppose that objects A and B have a product $A \times B$. Take any arrow $f : A \longrightarrow B$, the *graph of* f is defined as the arrow Γ_f that makes the following diagram commute:

$$\begin{tikzcd} & A \arrow[dl, "id_A"'] \arrow[d, "\Gamma_f"] \arrow[dr, "f"] & \\ A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B \end{tikzcd}$$

Definition 2.2.10 (Diagonal)**.** For any object A, the *diagonal* $\Delta_A : A \longrightarrow A \times A$ is a graph of id_A.

$$\begin{tikzcd}
& A \arrow[dl, "id_A"'] \arrow[d, "\Delta_A"] \arrow[dr, "id_A"] & \\
A & A \times A \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & A
\end{tikzcd}$$

Exercise 2.2.11. Show that, if a category $\mathcal{C}$ has binary products, $\langle h, k\rangle = (h \times k) \circ \Delta_T$ for any $h : T \longrightarrow A$ and $k : T \longrightarrow B$.

Definition 2.2.12 (Binary Coproduct)**.** A *binary coproduct* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A + B$ of $\mathcal{C}$
- two projections: morphisms $\iota_1 : A \longrightarrow A + B$ and $\iota_2 : B \longrightarrow A + B$, such that given any object C and morphisms $f : A \longrightarrow C$, $g : B \longrightarrow C$, there is a unique morphism $\binom{f}{g} : A + B \longrightarrow C$ for which $\binom{f}{g} \circ \iota_1 = f$ and $\binom{f}{g} \circ \iota_2 = g$, i.e. the following diagram commutes.

$$\begin{tikzcd}
A \arrow[r, "\iota_1"'] \arrow[dr, "f"'] & A + B \arrow[d, dotted, "\binom{f}{g}"] & B \arrow[l, "\iota_2"] \arrow[dl, "g"] \\
& C &
\end{tikzcd}$$

Theorem 2.2.13. For any objects A and B, a binary coproduct of A and B is unique up to isomorphism if it exists.

Theorem 2.2.14. In a category $\mathcal{C}$ with binary coproducts, any object A is isomorphic to $\emptyset + A$ ($A \cong \emptyset + A$).

Theorem 2.2.15. For any objects A, B, $A + B \cong B + A$.

Proof. Reverse the arrows in the proofs of Theorem 2.2.3, Theorem 2.2.8, and Theorem 2.2.4, respectively. □

Exercise 2.2.16. Show that the coproduct constructions are associative: for any objects A, B, C, $A + (B + C) \cong (A + B) + C$

2.3 Pullbacks and Pushouts

Definition 2.3.1 (Pullback)**.** A *pullback* for a pair of morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$ in a category $\mathcal{C}$ is given by:

- an object $A \times_C B$ of $\mathcal{C}$
- a pair of morphisms $p_1 : A \times_C B \longrightarrow A$ and $p_2 : A \times_C B \longrightarrow B$ (called *projection arrows*) for which $f \circ p_1 = g \circ p_2$ such that given any two morphisms $h : T \longrightarrow A$ and $k : T \longrightarrow B$ in $\mathcal{C}$ for which $f \circ h = g \circ k$, there is a unique morphism $m : T \longrightarrow A \times_C B$ for which $p_1 \circ m = h$ and $p_2 \circ m = k$, i.e. the following diagram commutes.

$$\begin{tikzcd}
T \arrow[dr, dotted, "m"] \arrow[drr, bend left, "k"] \arrow[ddr, bend right, "h"'] & & \\
& A \times_C B \arrow[r, "p_2"] \arrow[d, "p_1"'] & B \arrow[d, "g"] \\
& A \arrow[r, "f"'] & C
\end{tikzcd}$$

Theorem 2.3.2. A pullback for a given pair of morphisms is determined up to isomorphism.

Proof. Suppose that $D, d_A : D \longrightarrow A, d_B : D \longrightarrow B$ and $E, e_A : E \longrightarrow A, e_B : E \longrightarrow B$ are pullbacks for a pair of morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$. By the definition of pullback, there is a unique arrow from D to E and vice versa.

$$\begin{tikzcd}
 & E \arrow[dd, "e_A"] \arrow[dr, "e_B"] \arrow[dl, dotted, bend right, "!"'] & \\
D \arrow[rr, "d_B"'] \arrow[dr, "d_A"'] \arrow[ur, dotted, "!"'] & & B \arrow[d, "g"] \\
 & A \arrow[r, "f"'] & C
\end{tikzcd}$$

On the other hand, since D is a pullback, the arrow from D to D that makes the following diagram commute is unique, that is, id_D.

$$\begin{tikzcd}
D \arrow[dr, dotted, "id_D"] \arrow[drr, bend left, "d_B"] \arrow[ddr, bend right, "d_A"'] & & \\
 & D \arrow[r, "d_A"'] \arrow[d, "d_B"'] & B \arrow[d, "g"] \\
 & A \arrow[r, "f"'] & C
\end{tikzcd}$$

This means that the arrow $D \xrightarrow{!} E \xrightarrow{!} D$ is id_D. Similarly, the arrow $E \xrightarrow{!} D \xrightarrow{!} E$ is id_E. Hence $D \cong E$. $\square$

Exercise 2.3.3. Show that, for any arrow $f : A \longrightarrow B$ and object C, if A and B have a product with C, then the following square is a pullback.

$$\begin{tikzcd}
A \times C \arrow[r, "f \times id_C"] \arrow[d, "\pi_1"'] & B \times C \arrow[d, "\pi_1"] \\
A \arrow[r, "f"'] & B
\end{tikzcd}$$

Theorem 2.3.4. If the following diagram is a pullback and g is monic, then f' is also monic.

$$\begin{tikzcd}
\cdot \arrow[r, "f'"] \arrow[d, "g'"'] & \cdot \arrow[d, "f"] \\
\cdot \arrow[r, tail, "g"'] & \cdot
\end{tikzcd}$$

Proof. For any two arrows m, n such that $f' \circ m = f' \circ n$, we have $f \circ f' \circ m = f \circ f' \circ n$.

$$\begin{tikzcd}
{} \arrow[dr, shift left, "n"] \arrow[dr, shift right, "m"'] & & \\
& {} \arrow[r, "f'"] & {} \arrow[d, "f"] \\
& & {}
\end{tikzcd}$$

On the other hand, $f \circ f' = g \circ g'$. So we also have $g \circ g' \circ m = g \circ g' \circ n$. This means that the following diagram commutes.

$$\begin{tikzcd}
{} \arrow[dr, shift left, "n"] \arrow[dr, shift right, "m"'] \arrow[drr, bend left, "f' \circ m = f \circ n"] \arrow[ddr, bend right, "g' \circ m = g' \circ n"'] & & \\
& {} \arrow[r, "f'"] \arrow[d, "g'"'] & {} \arrow[d, "f"] \\
& {} \arrow[r, "g"'] & {}
\end{tikzcd}$$

Since the square is a pullback, $m = n$. ☐

Theorem 2.3.5. In the following diagram, the outer rectangle is a pullback if the two inner squares are pullbacks.

$$\begin{tikzcd}
{} \arrow[r] \arrow[d] & {} \arrow[r] \arrow[d] & {} \arrow[d] \\
{} \arrow[r] & {} \arrow[r] & {}
\end{tikzcd}$$

Proof. Consider any two arrows that make the following diagram commute.

$$\begin{tikzcd}
{} \arrow[drrr, bend left] \arrow[ddr, bend right] & & & \\
& {} \arrow[r] \arrow[d] & {} \arrow[r] \arrow[d] & {} \arrow[d] \\
& {} \arrow[r] & {} \arrow[r] & {}
\end{tikzcd}$$

Since the right square is a pullback, there is a unique arrow to the upper left corner of it.

$$
\begin{tikzcd}
{} \arrow[drr, dotted] \arrow[drrr, bend left] \arrow[ddr, bend right] & & & \\
& & {} \arrow[r] \arrow[d] & {} \arrow[d] \\
& {} \arrow[r] & {} \arrow[r] & {}
\end{tikzcd}
$$

Then, since the left square is also a pullback, there is a unique arrow to the upper left corner of it. Therefore, the whole square is a pullback.

$$
\begin{tikzcd}
{} \arrow[dr, dotted] \arrow[drr, dotted] \arrow[drrr, bend left] \arrow[ddr, bend right] & & & \\
& {} \arrow[r] \arrow[d] & {} \arrow[r] \arrow[d] & {} \arrow[d] \\
& {} \arrow[r] & {} \arrow[r] & {}
\end{tikzcd}
$$

$\square$

Exercise 2.3.6. Suppose that there is a commutative triangle

$$
\begin{tikzcd}
C \arrow[rr, "h"] \arrow[dr, "m"'] & & D \arrow[dl, "n"] \\
& B &
\end{tikzcd}
$$

and suppose that m and n have the following pullbacks along f:

$$
\begin{tikzcd}
E \arrow[r, "q"] \arrow[d, "p"'] & C \arrow[d, "m"] \\
A \arrow[r, "f"'] & B
\end{tikzcd}
\qquad
\begin{tikzcd}
F \arrow[r, "s"] \arrow[d, "r"'] & D \arrow[d, "n"] \\
A \arrow[r, "f"'] & B
\end{tikzcd}
$$

Show that, then, there is a unique $u : E \longrightarrow F$ that makes the following diagram commute, and E, u, q a pullback for h and s:

$$
\begin{tikzcd}
E \arrow[rrr, "q"] \arrow[drr, "u"] \arrow[ddr, "p"'] & & & C \arrow[drr, "h"] \arrow[ddr, "m"'] & & \\
& & F \arrow[rrr, "s"] \arrow[dl, "r"] & & & D \arrow[dl, "n"] \\
& A \arrow[rrr, "f"'] & & & B &
\end{tikzcd}
$$

2.4 Equalizers and Coequalizers

Definition 2.4.1 (Equalizer)**.** A morphism $e : E \longrightarrow A$ is an *equalizer* for a pair of morphisms $f, g : A \longrightarrow B$ in a category $\mathcal{C}$ iff

- $f \circ e = g \circ e$
- given any $h : T \longrightarrow A$ with $f \circ h = g \circ h$, there is a unique $k : T \longrightarrow E$ for which $h = e \circ k$.

i.e. the following diagram commutes.

$$\begin{tikzcd} E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\ T \arrow[u, dotted, "k"] \arrow[ur, "h"'] & & \end{tikzcd}$$

Theorem 2.4.2. Every equalizer is monic.

Proof. Suppose that e is an equalizer of f and g in the following diagram, i.e. $f \circ e = g \circ e$.

$$\begin{tikzcd} E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \end{tikzcd}$$

Consider two morphisms h and k such that $e \circ h = e \circ k$.

$$\begin{tikzcd} E \arrow[r, "e"] & A \\ T \arrow[u, shift left, "h"] \arrow[u, shift right, "k"'] \arrow[ur, "e\circ h=e\circ k"'] & \end{tikzcd}$$

Since $f \circ e = g \circ e$, it follows that $f \circ e \circ h = g \circ e \circ h$. Then there is a unique arrow from T to E that makes the following diagram commute.

$$\begin{tikzcd} E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\ T \arrow[u, dotted] \arrow[ur, "e\circ h"'] & & \end{tikzcd}$$

Since $e \circ h = e \circ k$, it turns out that $h = k$. Therefore e is monic. $\square$

Theorem 2.4.3. If a category has all binary products and equalizers for any two morphisms, then it has a pullback for any two morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$.

Exercise 2.4.4. Prove Theorem 2.4.3 by showing that the equalizer for $f \circ \pi_1$ and $g \circ \pi_2$ gives a pullback diagram for $f : A \longrightarrow C$ and $g : B \longrightarrow C$.

$$\begin{tikzcd}
T \arrow[dr, tail, "e"] & & B \arrow[dd, "g"] \\
& A \times B \arrow[ur, "\pi_2"] \arrow[dl, "\pi_1"'] & \\
A \arrow[rr, "f"'] & & C
\end{tikzcd}$$

Exercise 2.4.5. Suppose that a category has a pullback for every pair $f : A \longrightarrow C$ and $g : B \longrightarrow C$ and all binary products. Then it has an arrow from $A \times_C B$ to $A \times B$ and it is an equalizer for $f \circ \pi_1$ and $g \circ \pi_2$.

Theorem 2.4.6. The graph of $f : A \longrightarrow B$ is an equalizer for $f \circ \pi_1 : A \times B \longrightarrow B$ and $\pi_2 : A \times B \longrightarrow B$.

Exercise 2.4.7. Prove Theorem 2.4.6.

2.5 Kernel and Cokernel Pairs

Definition 2.5.1 (Kernel Pair)**.** For any morphism $f : A \longrightarrow B$, a *kernel pair* for f is the pair of projection arrows in a pullback of f along itself.

$$\begin{tikzcd}
K \arrow[r, "h"] \arrow[d, "k"'] & A \arrow[d, "f"] \\
A \arrow[r, "f"'] & B
\end{tikzcd}$$

The dual notion of a kernel pair is a *cokernel pair*.

Definition 2.5.2 (Cokernel Pair)**.** For any morphism $f : A \longrightarrow B$, a *cokernel pair* for f is the pair of projection arrows in a pushout of f along itself.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] & B \arrow[d, "h"] \\ B \arrow[r, "k"'] & C \end{tikzcd}$$

Theorem 2.5.3. Kernel pairs for any morphism are determined up to isomorphism.

Proof. It follows directly from Theorem 2.3.2. ☐

Theorem 2.5.4. If a cokernel pair for a morphism f consists of a single morphism taken twice, then f is epic.

Proof. Suppose that the pair $x, x : B \longrightarrow C$ is a cokernel pair for $f : A \longrightarrow B$ and two morphisms $g, h : B \longrightarrow T$ are such that $g \circ f = h \circ f$. Then there must be a unique arrow from C to T (say u) and $g = u \circ x = h$. Therefore f is epic.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] & B \arrow[d, "x"] \arrow[ddr, bend left, "g"] & \\ B \arrow[r, "x"'] \arrow[drr, bend right, "h"'] & C \arrow[dr, dotted, "u"] & \\ & & T \end{tikzcd}$$

☐

2.6 Cones and Cocones

Definition 2.6.1 (Cone)**.** (McLarty, 1992, p.48) Consider a diagram D with an object A_i at each vertex i, and an arrow f_e at each edge e. A *cone over* D consists of an object C and morphisms $p_i : C \longrightarrow A_i$ to each vertex such that, for every edge e with morphism $f_e : A_i \longrightarrow A_j$ we have $f_e \circ p_i = p_j$.

Definition 2.6.2 (Limit)**.** A *limit* for the diagram D is a cone over D (an object C and morphisms p_i) which satisfy the following properties:

- For any cone (say, T and q_i) over D, there is a unique arrow $u : T \longrightarrow C$, such that $p_i \circ u = q_i$ for each vertex i.

The morphisms p_i are called *projection arrows.*

Example 2.6.3. Consider a diagram which has no vertices. Then a cone over it is a limit iff it is a terminal object.

$$\begin{tikzcd} T \arrow[r, dotted, "u"] & C \end{tikzcd}$$

Example 2.6.4. Consider a diagram which has two vertices and no edges.

$$A_1 \qquad\qquad A_2$$

Then a cone over it is a limit iff it is a product diagram.

$$\begin{tikzcd} & T \arrow[dl, "q_1"'] \arrow[d, dotted, "u"] \arrow[dr, "q_2"] & \\ A_1 & C \arrow[l, "p_1"] \arrow[r, "p_2"'] & A_2 \end{tikzcd}$$

Example 2.6.5. Consider a diagram which has three vertices and two edges.

$$\begin{tikzcd} & A_2 \arrow[d, "g"] \\ A_1 \arrow[r, "f"'] & A_3 \end{tikzcd}$$

Then a cone over it is a limit iff it is a pullback diagram.

$$\begin{tikzcd}
T \arrow[dr, dotted, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\
& C \arrow[r, "p_2"] \arrow[d, "p_1"'] & A_2 \arrow[d, "g"] \\
& A_1 \arrow[r, "f"'] & A_3
\end{tikzcd}$$

Example 2.6.6. Consider a diagram which has two vertices and two edges between them.

$$\begin{tikzcd}
A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B
\end{tikzcd}$$

Then a cone over it is a limit iff it is an equalizer diagram.

$$\begin{tikzcd}
C \arrow[r, "p_1"] & A_1 \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & A_2 \\
T \arrow[u, dotted, "u"] \arrow[ur, "q_1"'] & &
\end{tikzcd}$$

Cocone is the dual notion of a cone, and *colimits* are the dual notion of limits.

Definition 2.6.7 (Cocone)**.** (McLarty, 1992, p.50) A *cocone* on the diagram D is an object C together with arrows $q_i : A_i \longrightarrow C$ for each vertex i, such that for every edge e with arrow $f_e : A_i \longrightarrow A_j$ we have $q_j \circ f_e = q_i$.

Definition 2.6.8 (Colimit)**.** (McLarty, 1992, p.50) A *colimit* for the diagram D is a cocone on D, namely C with arrows p_i, with the following property: given any cocone T, $r_i : A_i \longrightarrow T$ on D, there is exactly one arrow $u : C \longrightarrow T$ such that for every vertex $i, u \circ p_i = r_i$.

2.7 All Finite Limits and Colimits

Definition 2.7.1 (Finite Products)**.** A category $\mathcal{C}$ has *finite* products iff it has products for every finite diagram.

Definition 2.7.2 (All Finite Limits)**.** A category $\mathcal{C}$ has *all finite limits* iff $\mathcal{C}$ has a limit for every finite diagram.

Theorem 2.7.3. If a category has a terminal object, binary products, and an equalizer for every parallel pair of arrows, then it has all finite limits.

Proof. Suppose that there is a product $\Pi_i A_i$ of all the A_i, with projection $p_j : \Pi_i A_i \longrightarrow A_j$ for each vertex j. Use the notation A_e for the codomain of the arrow at edge e, so that if $f_e : A_j \longrightarrow A_k$ then $A_e = A_k$. Suppose that there is a product $\Pi_e A_e$ with one factor A_e for each edge e, and a projection $p_e : \Pi_e A_e \longrightarrow A_k$ for each $f_e : A_j \longrightarrow A_k$.

Let $r : \Pi_i A_i \longrightarrow \Pi_e A_e$ be the arrow such that, for each edge e with $f_e : A_j \longrightarrow A_k$, the following triangle commutes:

$$\begin{tikzcd}
\Pi_i A_i \arrow[r, "r"] \arrow[dr, "p_k"'] & \Pi_e A_e \arrow[d, "p_e"] \\
& A_k
\end{tikzcd}$$

Let $s : \Pi_i A_i \longrightarrow \Pi_e A_e$ be the arrow such that for each edge e the following square commutes:

$$\begin{tikzcd}
\Pi_i A_i \arrow[r, "s"] \arrow[d, "p_j"'] & \Pi_e A_e \arrow[d, "p_e"] \\
A_j \arrow[r, "f_e"'] & A_k
\end{tikzcd}$$

Then the diagram D has a limit iff there is an equalizer for r and s. More specifically, $l : L \longrightarrow \Pi_i A_i$ is an equalizer for r and s iff L together with all $p_i \circ l : L \longrightarrow A_i$ forms a limit cone for D.

Consider any cone $T, h_i : T \longrightarrow A_i$ over D. There is a unique $h : T \longrightarrow \Pi_i A_i$ such that for each vertex k we have $p_k \circ h = h_k$. Then $r \circ h = s \circ h$ iff for each edge

e, $p_e \circ r \circ h = p_e \circ s \circ h$. But, by the definition of r, s and h, that last equation means that $h_k = f_e \circ h_j$, where $f_e : A_j \longrightarrow A_k$. Thus T, h factors uniquely through L, l iff the cone T, h_i factors uniquely through the cone $L, p_i \circ l$. □

Theorem 2.7.4. If a category has a terminal object and pullbacks for all corner of arrows, then it has all finite limits.

Proof. By Theorem 2.7.3, Theorem 2.7.5 and Theorem 2.7.7. □

Theorem 2.7.5. A category with a terminal object and pullbacks for all corners of arrows, has a product for each pair of objects.

Exercise 2.7.6. Prove Theorem 2.7.5 by showing that $\langle A \times_1 B, p_1, p_2 \rangle$ in the following diagram is also a product diagram for A and B.

$$\begin{tikzcd}
A \times_1 B \arrow[r, "p_2"] \arrow[d, "p_1"'] & B \arrow[d, "!_B"] \\
A \arrow[r, "!_A"'] & 1
\end{tikzcd}$$

Theorem 2.7.7. A category with a product for each pair of objects, and a pullback for every corner of arrows, has an equalizer for every parallel pair.

Exercise 2.7.8. Prove Theorem 2.7.7 by showing that the following e is an equalizer for any pair $f, g : A \longrightarrow B$.

$$\begin{tikzcd}
E \arrow[r, "p_2"] \arrow[d, tail, "e"'] & B \arrow[d, tail, "\Delta_B"] \\
A \arrow[r, "\langle f,g \rangle"'] & B \times B
\end{tikzcd}$$

Bibliography

Crole, R. L. (1993) *Categories for Types.* Cambridge University Press.

Kleisli, H. (1965) “Every standard construction is induced by a pair of adjoint functors”, In the Proceedings of *the American Mathematical Society*, Vol. 16. pp.544–546.

MacLane, S. (1997) *Categories for the Working Mathematician*, Graduate Texts in Mathematics. Springer, second edition edition.

MacLane, S. and I. Moerdijk. (1992) *Sheaves in Geometry and Logic – A First Introduction to Topos Theory –*. Springer.

McLarty, C. (1992) *Elementary Categories, Elementary Toposes*, Oxford Logic Guides 21. Oxford Science Publications.

Moggi, E. (1989) “Computational lambda-calculus and monads”, In the Proceedings of *Fourth Annual IEEE Symposium on Logic in Computer Science.* pp.14–23.

Pedicchio, M. C. and W. Tholen. (2004) *Categorical Foundations: Special Topics in Order, Topology, Algebra, and Sheaf Theory*, Vol. 97 of *Encyclopedia of Mathematics and Its Applications.* Cambridge University Press.